



The role of attitudes and attention in pro-environmental decision-making: An eye-tracking study

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ABSTRACT

Encouraging environmentally sustainable lifestyles demands a deeper understanding of the processes underlying pro-environmental decision-making. This study investigates the role of environmental attitudes and attentional processes using eye-tracking technology during a decision-making task that involves real trade-offs between personal financial rewards and environmental consequences. We found partial evidence that stronger environmental attitudes – derived from the Campbell paradigm, but not from biospheric values – predicted an attentional prioritization of environmental information (i.e., carbon emissions). Specifically, stronger Campbellian attitudes were associated with a higher proportion of fixation time on this information, and an increasing likelihood of fixating this information first once participants became familiar with the task. However, the former effect may partly reflect gender-related differences in ecological considerations. The attention-behavior link appears context-dependent, influenced by factors such as the magnitude of financial incentives and environmental stakes. Our findings suggest that, while attentional processes may play a significant role in less financially rewarding pro-environmental decisions, they do not predict those with high financial stakes. These insights underscore the importance of considering both attentional dynamics and decision context in developing strategies to promote pro-environmental choices.

1. Introduction

Global climate change poses a pressing challenge to humanity, with individual behavior recognized as a significant contributor (Ivanova et al., 2020). To encourage more sustainable lifestyles, it is crucial to advance our understanding of pro-environmental decision-making and the attitudes driving these behaviors (Gifford & Nilsson, 2014; Nielsen et al., 2024; van Valkengoed et al., 2022). Scientists have recently called for a stronger focus on the decision-making process involved in pro-environmental choices (e.g., Doell et al., 2023; Nielsen et al., 2021; Sawe & Chawla, 2021) to develop more predictive models and nuanced theories. This study aims to answer this call by enriching behavioral findings with eye-tracking data from a consequential environmental decision-making task, providing deeper insights into how environmental attitudes are linked to attentional focus and subsequent choices.

1.1. From environmental attitudes to behavior: the role of attention

Emerging research has been devoted to unraveling the complexity of pro-environmental decision-making, revealing numerous factors driving pro-environmental behaviors (e.g., Brosch & Steg, 2021; van Valkengoed et al., 2022). A significant focus has been put on understanding why environmental attitudes and intentions often fail to predict corresponding behaviors (e.g., Carrington et al., 2014; Farjam et al., 2019; Juvan & Dolnicar, 2014; Nielsen et al., 2022). However, while predicting choice outcomes remains an important research goal, there is a growing need to shift focus from *why* people act sustainably to *how* these decisions unfold. This shift is crucial, as a more nuanced understanding of the processes that bridge attitudes and behaviors aids in developing better models of pro-environmental behavior, designing more effective experiments, and enhancing policy interventions (Kononov & Ruff,

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2022; Krajbich et al., 2010; Schulte-Mecklenbeck et al., 2017). For example, a global study showed that although conservatives tend to exhibit weaker climate change beliefs and attitudes than liberals, both groups engaged similarly in a tree-planting task. Climate interventions, however, boosted beliefs, policy support, and action for liberals but had limited or even negative effects on conservatives' climate beliefs and actions (Berkebile-Weinberg et al., 2024). This highlights the importance of focusing on the mechanisms underpinning decision-making processes, as understanding how individuals gather, process, and prioritize information can provide critical insights into the factors that drive sustainable behaviors (e.g., Krajbich et al., 2010; Schulte-Mecklenbeck et al., 2019). By exploring why and how decisions are made, researchers can uncover why interventions resonate differently across populations and contexts (Kononov & Ruff, 2022; Schulte-Mecklenbeck et al., 2011, 2019), ultimately paving the way for more targeted and effective strategies to promote pro-environmental actions.

One approach in which attitudes can impact decisions is via attention (Fridman et al., 2018; Gwinn & Krajbich, 2020; Smith & Krajbich, 2019). There is growing evidence that attention plays a crucial role in shaping decisions across various domains (Amasino et al., 2019; Gwinn & Krajbich, 2020; Krajbich, 2019; Milosavljevic et al., 2012). Harnessing insights from attentional processes preceding choices are particularly valuable in predicting multi-faceted decisions made based on different information in varying contexts and by diverse individuals (Kononov & Ruff, 2022). In such decisions, individuals must process multiple attributes, weigh competing considerations, and prioritize relevant details, making understanding attentional allocation key to understanding choice outcomes. In pro-environmental decision-making, incorporating eye-tracking data provides an opportunity to capture such attentional processes. In fact, by detecting consecutive gaze positions in real time, this tool offers direct, non-intrusive measures of information acquisition and weighting before a decision is reached, enabling researchers to infer mechanisms of information gathering and cognitive processing (Ashby et al., 2016; Glaholt & Reingold, 2011).

1.2. Insights from eye-tracking in pro-environmental behavior research

Despite the significant potential of eye-tracking to elucidate pro-environmental decision-making, its application in the study of pro-environmental behavior has been limited. Sollberger et al. (2017) found that lower stress and stronger pro-environmental orientation were linked to longer visual attention towards climate change images. She and MacDonald (2018) observed that sustainability-triggering information on products attracted participants' attention and increased hypothetical choices of sustainable products. Recently, Gómez-Carmona et al. (2021) found that individuals more concerned with environmental issues tend to focus more quickly and frequently on negatively framed renewable energy ads.

Studies have also explored the impact of labels and visual information on consumer behavior. Brazil and Caulfield (2017) showed that well-designed energy labels with intuitive features, such as color coding and alphabetic grading, effectively captured consumer attention and improved information recall. Fujisawa et al. (2022) demonstrated that energy labels in real estate advertisements increased understanding of energy standards through repeated exposure and optimized design. Similarly, Waechter et al. (2016) found that shorter energy-efficiency scales enhanced consumers' perceptions of product differences and motivated energy-friendly choices. However, attention to eco-labels often varies based on context (e.g., Babakhani et al., 2020). For example, Grebitus and Van Loo (2022) found that pesticide-free labels attracted more visual attention and were linked to a higher willingness to pay (WTP), whereas labels indicating the absence of genetically modified organisms (GMO-free labels) received less attention and had a lower associated WTP. Similarly, Orquin et al. (2020) highlighted how competing visual elements influence label salience, which can dilute their impact on consumer decisions. While these studies demonstrate the

contextual variability of attention to eco-labels, they do not examine how individual differences, such as environmental attitudes, shape attentional processes in decision-making. Moreover, prior studies have mostly relied on self-reports or hypothetical choice scenarios, limiting our understanding of how attention unfolds in real and consequential decision-making situations (Lange et al., 2023). To address these gaps, our study explores how environmental attitudes influence attentional processes in pro-environmental decision-making. Using an abstract decision paradigm with real environmental consequences, we go beyond self-reports and hypothetical scenarios in an effort to provide novel insights into the attentional mechanisms underlying environmentally relevant decision-making.

1.3. Aim of the current study

Given the limited understanding of how individuals make pro-environmental choices and how these unfold over time, this research aims to explore the attentional processes underlying these choices. Specifically, we aim to address two research questions. First, are environmental attitudes linked to distinct patterns of search strategies, information weighting and processing? Second, are these gaze patterns indicative of pro-environmental choices? To answer these questions, we will examine eye movements during an environmental decision-making task involving repeated choices (adapted from Berger & Wyss, 2021). The task aims at tapping into trade-offs between unsustainable, personal rewards and long-term environmental benefits by presenting participants with repeated opportunities to accept or reject a financially rewarding, but environmentally harmful offer. Accepting the offer results in a monetary bonus and actual carbon dioxide (CO₂) emissions while rejecting it requires foregoing the bonus but staying CO₂-neutral. Carbon emissions are realized through the purchase and retirement of CO₂-certificates through the European Union Emission Trading System (EU-ETS; see methods for details). This task thus mirrors real-world trade-offs, such as deciding whether to book a rewarding yet carbon-intensive flight for a vacation, and therefore serves as an indicator of a person's pro-environmental propensity (Lange, 2024). Previous research has shown that people with stronger environmental attitudes are more likely to resolve such trade-offs in favor of the environment (e.g., Berger & Wyss, 2021; Bouman et al., 2021; de Groot & Thøgersen, 2018; Enzler et al., 2019; Wyss et al., 2022), especially when these attitudes are assessed using the Campbell paradigm developed by Kaiser et al. (2010) (e.g., Kaiser & Byrka, 2014; Kaiser & Lange, 2021). This paradigm derives the strength of an individual's attitude (referred as Campbellian attitude measures) in the degree of difficulty they overcome to achieve the goal implied by that attitude.

To comprehensively assess environmental attitudes, we incorporate both conventional self-reported measures [Schwartz value scale (SVS); Steg & de Groot, 2012] and the Campbell Paradigm (Kaiser et al., 2010). While traditional measures capture explicit beliefs and values, the Campbell Paradigm conceptualizes attitudes as latent dispositions inferred from self-reported behavioral tendencies. Using the Rasch model, this approach structures behaviors along a continuum of difficulty, estimating attitude strength based on an individual's willingness to engage in progressively costlier pro-environmental actions (from publicly expressing support for environmental protection to refraining from air travel). This probabilistic framework provides a behaviorally grounded measure of attitudes, distinct from observed behavior, which can be influenced by situational constraints. By integrating both approaches, we acknowledge the ongoing debate about attitude conceptualization (Ajzen, 2001; Eagly & Chaiken, 1993) and examine whether decision-making processes align more closely with explicit self-reports or cost-sensitive behaviorally inferred attitudes.

1.4. Hypotheses

Given the scarcity of eye-tracking studies in pro-environmental

behavior research, we developed our hypotheses based on the extensive body of research using eye-tracking as a process tracing method in social decision-making (see [Rahal & Fiedler, 2019](#) for a review). Numerous studies have shown that the proportion of attention paid to specific information reflects its relative importance ([Glöckner & Herbold, 2011](#); [Hu et al., 2020](#); [Tanida & Yamagishi, 2010](#)). In other words, the relative dwell time - defined as the proportion of fixation time directed toward a specific area of interest (AOI) relative to the overall fixation time - indicates the importance of specific aspects of the decision-making context, with more important AOIs receiving longer attention (e.g., [Fiedler et al., 2013](#); [Polonio et al., 2015](#)). For instance, research in prosocial decision-making shows that participants with stronger individualistic social values focus more on their own pay-off than on others' pay-offs in money allocation tasks (e.g., [Fiedler et al., 2013](#); [Rahal et al., 2020](#)). Translating these findings to the context of pro-environmental decision-making, we expect that individuals' environmental attitudes will predict the extent to which they devote attention to environmental information (e.g., carbon footprint, resource consumption) compared to rewarding information (e.g., price discounts, bonuses). Specifically, we hypothesize (H1) that stronger environmental attitudes (assessed by conventional or Campbellian measures) will be associated with a higher relative dwell time on the environmental attribute (i.e., carbon emissions) during decision-making.

In addition to the proportion of attention allocated to a specific attribute, another category of attention measures examines the timing of fixations (i.e., *when* during the process a particular attribute is fixated). One such measure is the first fixation on an AOI, which identifies where the initial gaze lands at the start of visual inspection. This measure captures early attention ([Todd et al., 2016](#)) and may be influenced by pre-existing preferences or goals, especially when AOI locations are consistent across trials ([Glaholt & Reingold, 2009](#); [Gwinn et al., 2019](#)). For example, [Werthmann et al. \(2011\)](#) demonstrated that overweight participants initially fixated on pictures containing high-fat food cues more frequently than healthy-weight participants. Accordingly, we hypothesize (H2) that stronger environmental attitudes will predict a higher likelihood of directing the first fixation towards the environmental attribute.

Beyond examining whether environmental attitudes are associated with distinct attentional processes, we also aim to investigate whether these attentional patterns are linked to actual behavioral outcomes. Numerous studies have demonstrated that the proportion of fixation time during the decision-making process is correlated with subsequent choice outcomes ([Jiang et al., 2016](#); [Polonio et al., 2015](#); [Wei et al., 2022](#)). For instance, in the realm of prosocial decision-making, [Peshkovskaya and Myagkov \(2020\)](#) found that longer dwell times on cooperative choice options are correlated with more cooperative decisions. Thus, we hypothesize (H3) that higher relative dwell time on the environmental attribute will be associated with a higher likelihood of making a pro-environmental choice (i.e., rejecting a financially beneficial but environmentally harmful offer).

In addition to relative dwell time, insights about information weighting can be inferred from the last fixation before a choice is made, as the last-seen AOI has been shown to predict subsequent choices across various contexts (e.g., [Ghaffari & Fiedler, 2018](#); [Liu et al., 2020](#); [Stewart et al., 2016](#)). In other words, one can utilize the last fixation to determine the information that is essential for crossing the threshold from the decision-making process to the actual decision ([Rahal & Fiedler, 2019](#)). Consequently, we expect (H4) that last fixations on the environmental attribute will be positively associated with pro-environmental decisions.

After examining the link between attentional measures and pro-environmental decisions in H3 and H4, we aim to assess whether attention acts as a moderator in the relationship between pro-environmental attitudes and behaviors. Evidence for this possibility comes from prior findings in risky decision-making research ([Hoven et al., 2023](#)) and consumer reactions to advertisements ([Amaro et al., 2023](#)), which suggest that attention can amplify or dampen the impact

of attitudes on behavior. More specifically, we investigate whether the weighting and relative importance allocated to the environmental attribute in the decision-making process (measured by the relative dwell time, first fixation location, last fixation location) moderate the association between attitudes and decisions. Second, we explore whether the attitude-behavior relationship is moderated by the processing depth and effort of information integration made before reaching the decision ([Barrafrem & Hausfeld, 2020](#); [Rahal & Fiedler, 2019](#)). Previous research has linked higher-level deliberate processing in decision tasks to longer fixation durations on outcomes ([Glöckner & Herbold, 2011](#); [Rubaltelli et al., 2023](#)) and more frequent information reinspection ([Horstmann et al., 2009](#); [Wang et al., 2024](#)). Thus, we used eye-tracking measures (i.e., mean fixation duration, number of transitions between attributes) to test the moderating role of the depth of deliberation on the attitude-behavior relationship. The procedure, hypotheses and analyses were preregistered using the Open Science Framework (<https://doi.org/10.17605/OSF.IO/FBG4Z>, registered in April 2024).

2. Material and methods

2.1. Participants

To determine the required sample size, we performed a simulation-based power analysis using parameters from a pilot study ($N = 15$) with the same design and procedure. Power analysis results (based on 1000 simulations using the SIMR package in R; [Green & MacLeod, 2016](#)) indicated that a sample size of 100–120 participants would provide 95 % power to detect the effect sizes estimated in the pilot study with an alpha of 0.05. These effect sizes were estimated for the mixed-effects models testing H1 and H3 (see [Figs. S1 and S2](#) in supplementary material). Additionally, a sample size greater than $N = 100$ is required to calibrate the Rasch scale used in Campbell's paradigm for accurate cost estimates ([Wright, 1977](#)). As preregistered, we aimed for a sample size of $N = 110$ and collected data from 115 participants to account for potential exclusions.

We recruited 115 university student participants based on the following inclusion criteria: age between 18 and 40, absence of psychological and neurological disorders, no regular intake of psychotropic drugs, normal or corrected-to-normal vision, German speakers. Participants were asked to refrain from wearing eye make-up and give preference to contact lenses in case of corrected vision. In addition, we excluded students of psychology, economics, and social sciences since they might have come across comparable experiments during their studies and therefore act differently from naïve participants, and students of biology since they might be more sensitive to environmental issues and act more pro-environmentally than average.

2.2. Measures

2.2.1. Environmental decision-making task

We used an adapted version of the carbon emission task ([Berger & Wyss, 2021](#)) in which we presented 36 financial rewards paired with actual carbon emissions that participants could either accept (i.e., unsustainable choice) or reject (i.e., sustainable choice). The offers comprised seven levels of financial rewards and carbon emissions (see [Fig. 1](#)). To make information about carbon emissions more accessible, participants were provided in the instructions with equivalents of CO₂ in terms of car journeys, days spent heating a flat and number of meat burgers. Participants were told that, at the end of the experiment, one of the trials will be randomly selected and that the decision made by the participant for this trial will be implemented. In other words, if participants had rejected the selected offer, they would not earn the additional payment, and the amount of CO₂ attached to the option would be acquired in form of a carbon emission certificate from the EU-ETS using the service provider [compensators.org](#). The EU-ETS regulates the carbon emissions produced by significant polluters within the European Union,

	5 CHF	8 CHF	11 CHF	14 CHF	17 CHF	20 CHF	23 CHF
20kg CO ₂							
35kg CO ₂							
50kg CO ₂							
65kg CO ₂							
80kg CO ₂							
95kg CO ₂							
110kg CO ₂							

Fig. 1. Overview of trials based on carbon emission and monetary bonus levels. The trials that were presented to participants are filled in blue or yellow. Blue squares include relatively high levels of carbon emissions paired with relatively low levels of financial incentives (i.e., carbon level > bonus level) and yellow squares include relatively low levels of carbon emissions paired with relatively high levels of financial incentives (i.e., carbon level < bonus level). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

such as airlines and energy companies. Buying and “retiring” certificates from the EU-ETS helps to lower the total global emissions limit and has become a common method to allocate real environmental consequences to experimental behavior (Ockenfels et al., 2020; Wyss et al., 2021). On the other hand, for all selected offers that had been accepted, the carbon certificates remained in the market and could be used for pollution by the firms.

To ensure behavioral variance while keeping the task brief, we avoided a fully crossed design with 49 offers and presented 36 offers to participants in a quasi-randomized order (Fig. 1). We quasi-randomly assigned these offers to 6 blocks. Each block contained 3 offers that included relatively high levels of carbon emissions paired with relatively low levels of financial incentives, and 3 offers that include relatively low levels of carbon emissions paired with relatively high levels of financial incentives (see Fig. 1). This quasi-randomized presentation aimed to ensure a similar distribution of offers within and among participants. The order of blocks and trials within blocks was randomly determined for each participant.

2.2.2. Questionnaires

We used the General Ecological Behavior (GEB) scale, a well-established instrument with 50 self-reported ecological behaviors, as a Campbell-paradigm-based measure of environmental attitudes (Kaiser et al., 2010; Kaiser & Lange, 2021; Kaiser & Wilson, 2019). This approach employs the dichotomous Rasch model to scale behaviors based on their difficulty in our sample and estimate participants’ attitude score (in logits) from the ratio of the probability of engagement/non-engagement across items with increasing levels of behavioral costs (higher logits reflecting higher attitude score). This instrument has been shown to represent a reliable and valid measure of environmental attitude (Kaiser et al., 2010; Lange & Dewitte, 2019). Items were slightly amended from Kaiser and Lange (2021) for our Swiss student sample (see Supplementary Table S1 for item details and cost estimates) and rated on a 5-point frequency scale (from 1 = never to 5 = always) or a yes-no format (“not applicable” responses were treated as missing values). The person separation reliability was acceptable ($rel = 0.78$) to distinguish between different levels of environmental attitude in our sample. The psychometric features of the instrument indicated a good model fit, with a mean infit mean square value ranging between the threshold values of 0.7 and 1.3 ($MS = 0.98$, $SD = 0.28$), a standardized fit statistic $|t|$ lower than 1.96 ($M_{infit} = -0.15$, $SD_{infit} = 1.29$) and only 3.48 % of participants exhibiting misfit (Wright & Linacre, 1994). We used the biospheric value orientations subscale of the SVS

(Steg & de Groot, 2012) as a conventional measure of environmental attitudes. The four items were rated on a Likert-scale from –1 “opposed to my values” to 7 “extremely important”. The internal reliability of the biospheric subscale was good ($\alpha = .88$). Finally, participants were asked to rate how effective they believe EU-ETS to be in reducing greenhouse gases on a 5-point scale from “not at all effective” to “very effective”.

2.3. Procedure

After receiving a thorough explanation of the procedure and providing written informed consent, participants were tested in a laboratory room without natural light at the university, seated on a desk chair in front of a computer and a screen-based eye-tracker camera. Following a 9-point eye-gaze calibration and 5-point validation, they completed four practice trials and had to correctly answer control questions to confirm their understanding of the task before starting the experimental trials. Throughout the task, eye-tracking data was recorded.

At the beginning of each trial (Fig. 2), a black fixation dot appeared centrally on a grey background screen for 1000ms to ensure an initial focus of attention at the center of the screen. The fixation dot also serves as a drift check at the beginning of each block to confirm the reliability of the calibration and maximize data quality throughout the task. The environmental and financial attributes of the offer were then displayed on the upper and lower halves of the screen (counterbalanced by subjects and gender; see Hu et al., 2020 for similar procedure). Participants indicated their decision by pressing F to accept the offer (i.e., unsustainable decision), or J to reject the offer (i.e., sustainable decision), with responses being marked by a black or white sticker on the keyboard. They had maximum 15 s to decide, and the offer remained on the screen until a response. At the end of each trial, decision deliberation was assessed by asking participants to which extent they had to think to make the decision on a 3-point Likert scale from “barely at all” to “rather a lot”.

At the end of the task, participants filled in questionnaires assessing their environmental attitudes and were then informed which offer was randomly selected. As compensation, participants received a fixed fee of CHF 20.- for participating in the study, along with an additional payment based on their individual decisions (ranging from CHF 0.- to CHF 23.-). The experiment lasted around 30 min. Our study adheres to all applicable ethical regulations since it has been approved by the local ethics committee (approval no. 2023-08-01) and was conducted in accordance with the declaration of Helsinki.

2.4. Eye-tracking recording and processing

Participants were placed 60 cm away from a desktop PC with a 21.5” LED screen (resolution 1920x1080, refresh rate of 60Hz). We recorded eye movements using a Tobii Pro Spark (video-based pupil and corneal reflection eye-tracker, sampling rate of 60Hz, average accuracy of 0.45°, root-mean-square precision of 0.26°, binocular). We controlled the presentation of the experimental task using OpenSesame (Mathôt et al., 2012) and developed a custom Python software for its synchronization with the eye-tracker (available as an open-source project on GitHub; see <https://github.com/humdek-unibe-ch/tbi-gaze>). The calibration and validation procedure were systematically restarted when accuracy was greater than 1.5°.

Gaze data was clustered into fixations with Dispersion-Based Identification algorithm (Salvucci & Goldberg, 2000). A fixation was defined as a cluster of single measurements appearing within a minimum time of 100ms and a 0.5° threshold for the dispersion between points. We defined two non-overlapping rectangular AOIs around the environmental and financial attributes that matched on color (black font on grey background), size (384x216 pixels or 9.1 × 5.1°) and quantity of content. The distance between the center of the two AOIs was 9.4°, which ensured to record the object of regard unambiguously and precluded

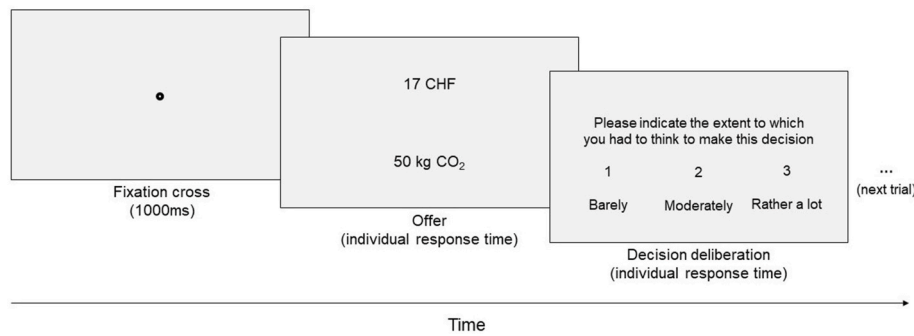


Fig. 2. Trial procedure and timing of the environmental decision-making task.

peripheral identification of the adjacent information while keeping the attentional cost of gaze transitions relatively low (Orquin & Mueller Loose, 2013; Schulte-Mecklenbeck et al., 2019). For the analyses, fixations outside AOIs were not considered.

2.5. Data analysis

2.5.1. Data pre-processing

A data reduction procedure was performed by removing trials in which (1) more than half of the fixations were allocated outside the AOIs, (2) decision time was shorter than 200ms, and (3) decision time was longer than 3 standard deviations above the mean. We also excluded participants if more than half of their trials were invalid (following the above procedure or due to poor calibration). Following this procedure, one participant was removed from the analyses, as were 136 invalid trials from the remaining sample (3.31 % of trials). All eye-tracking indices (i.e., relative dwell time, first fixation location, last fixation location, mean fixation duration and number of gaze transitions) towards AOIs were aggregated per trial. The data was prepared, aggregated, and analyzed using R (R Core Team, 2017).

2.5.2. Data analyses

All analyses were pre-registered, and exploratory analyses are explicitly labelled as such. We used linear mixed-effects models (LMMs) or generalized linear mixed-effects models (GLMMs) depending on the nature of the dependent variable (continuous or binary outcomes), with random intercepts for participants and trials unless otherwise specified.

We tested H1 using LLMs with the relative dwell time (calculated as the ratio of dwell time on the environmental AOI to the combined dwell time on both the environmental and financial AOIs) as the dependent variable, and environmental attitudes as a fixed-effect predictor, separately for Campbellian attitudes (scored in logits) and SVS attitudes (score range between -1 and 7). Model 1 included environmental attitudes and attribute position (dummy-coded 1 for carbon emissions at the bottom, 0 for carbon emissions at the top) as fixed effects. Model 2 added gender (dummy-coded 1 for women, 0 for men), age, perceived EU-ETS efficacy, and decision consistency as covariates to test the robustness of the effect.

We tested H2, H3 and H4 using GLMMs. For H2, we included the first fixation (dummy-coded 1 if on the environmental AOI, coded 0 if on the financial AOI) as dependent variable, and environmental attitudes (Campbellian or SVS) as a fixed-effect predictor. For H3, we included pro-environmental decision (dummy-coded 1 for declined offers, 0 for accepted) as dependent variable, and the relative dwell time on the environmental AOI as the fixed-effect predictor. For H4, we included pro-environmental decision as dependent variable, and the last fixation (dummy-coded 1 if on the environmental AOI, 0 if on the financial AOI) as the fixed-effect predictor. For each hypothesis, Model 3 included the main predictor and attribute position as fixed effects. Model 4 added gender, age, EU-ETS efficacy, and decision consistency as covariates to test the robustness of the effect. For H3 and H4, we also ran a non-

preregistered simplified model including only a random intercept for participants.

As preregistered, we explored whether attentional processes moderate the relationship between environmental attitudes and decisions using GLMMs (with random intercepts for participants and trials) separately for each eye-tracking measure (i.e., relative dwell time, first fixation, last fixation, mean fixation, number of gaze transitions), with interaction terms and attribute position as fixed effects (Model 3). When the interaction was significant, we performed sensitivity analyses controlling for the other covariates to test their robustness (Model 4).

Attribute position on the screen (based on the parity of participants' study number) was included as a covariate in all models since an attentional bias towards the attribute on the upper visual field (known as superior bias; Holmqvist et al., 2011; Strauch et al., 2024) was observed across all eye-tracking measures.² This adjustment was made despite preregistration plans to include attribute position only in models with first and last fixation measures. Decision consistency, used as a covariate in Models 2 and 4, was computed as the absolute difference between accepted and declined decisions, divided by the total number of trials. Scores range from 0 to 1, where 0 indicates maximally diverse decisions with half of the offers accepted, and 1 indicates complete consistency with all offers either accepted or declined.

Before interpreting the results, we examined multicollinearity by checking variance inflation factors (VIF). For each hypothesis, model comparisons were performed between Model 1 and Model 2 (for LMMs) and between Model 3 and Model 4 (for GLMMs) using Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and likelihood ratio tests (LRT) calculated from maximum likelihood parameter estimates. Based on these criteria, we reported and interpreted results from each model favored by at least one criterion, while full outputs of all tested models are provided in the Supplementary Material. Post-hoc simple slopes analyses and graphs were created for the most fitting models to interpret interaction effects. All models with interaction terms used centered predictor variables.

3. Results

3.1. Descriptive statistics

Our final sample of $n = 114$ participants consisted of 69 women and

² Participants who were presented with carbon emissions at the top of the screen showed a higher proportion of first fixations ($t(112) = 28.89, p < .001$) and last fixations ($t(112) = 3.80, p < .001$), as well as a higher relative dwell time ($t(112) = 6.91, p < .001$) on this attribute compared to participants who were presented with carbon emissions at the bottom of the screen. These findings are in line with Glaholt et al. (2010) who tested the effect of option location on attention and choice and found greater fixation duration and number of fixations for option located on the top compared to bottom (while no effect of option location was found on decision).

Table 1

Descriptive statistics of the independent (IV) and dependent (DV) variables.

Variables	IV or DV	Mean or Ratio	Std. Dev.	Min	Max
Attribute position (carbon on top/bottom)	IV	58/56	–	–	–
Gender (W/M)	IV	69/45	–	–	–
Age (in years)	IV	21.72	2.85	18	35
Decision consistency	IV	0.48	0.31	0	1
Perceived EU-ETS efficacy	IV	2.75	0.87	1	5
Campbellian attitude (in logits)	IV	0.63	0.86	-1.57	2.33
SVS attitude	IV	4.64	1.38	0.5	7
Last fixation on carbon (proportion)	IV	0.51	0.19	0.08	0.97
Mean fixation duration (in ms)	IV	236	30	176	319
Number of transitions	IV	3.10	1.03	0.46	5.66
Relative dwell time on carbon	IV, DV	0.54	0.14	0.15	0.82
First fixation on carbon (proportion)	IV, DV	0.52	0.42	0	1
Pro-environmental decision (proportion)	DV	0.65	0.25	0	1

45 men, with a mean age of 22 years ($SD = 2.8$; see Table 1). Participants showed relatively high levels of pro-environmental decisions (65 % of their decisions). Among them, 9 participants made all their decisions pro-environmental, while 3 participants never made any pro-environmental decisions. The mean decision consistency of participants was 0.48 ($SD = 0.31$). Their Campbellian attitude score fell between -1.57 and 2.33 logits ($M = 0.63$, $SD = 0.86$) and their SVS attitude score fell between 0.5 and 7 ($M = 4.64$, $SD = 1.38$).

3.2. Main analyses

3.2.1. Environmental attitudes predicting relative dwell time on environmental attribute

To test the effect of environmental attitude on the relative dwell time on environmental attribute (H1), we fitted LMMs including random intercepts for participants and trials, while controlling for attribute position in Model 1 and all covariates in Model 2.

For models including Campbellian attitude as a predictor, multicollinearity was low, with a maximum VIF of 1.12. Model comparisons yielded mixed results: while the LRT ($\chi^2(4) = 10.77$, $p = .029$) and

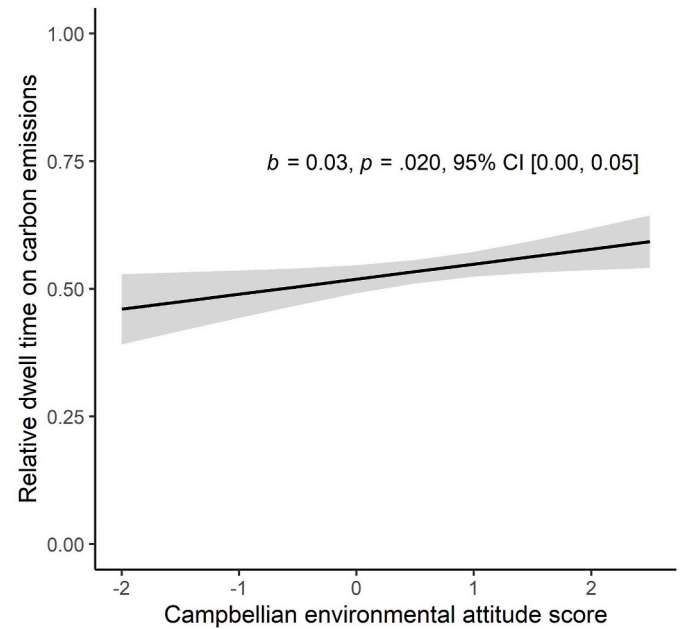


Fig. 3. Adjusted marginal effects predictions for relative dwell time on carbon emissions as a function of Campbellian attitude score (measured in logits) in Model 1 (adjusted for attribute position).

difference in AIC values ($\Delta AIC = 3$) favored Model 2, the difference in BIC values ($\Delta BIC = 22$) favored Model 1, reflecting its more conservative approach to model specification and parsimony (see Table 2). We therefore report both models to provide a comprehensive overview of findings and interpret covariate effects with appropriate caution. The hypothesis was supported by Model 1. The fixed effect of Campbellian attitude was significantly positive ($b = 0.03$, $p = .020$, 95 % CI [0.00–0.05]), indicating that stronger Campbellian attitude was associated with an increased proportion of attention paid to carbon emission relative to monetary information (see Table 2 and Fig. 3). The effect was attenuated and became marginal in Model 2 when accounting for all covariates ($b = 0.02$, $p = .085$, 95 % CI $[-0.00 - 0.05]$), suggesting a potential confounding effect of gender. Further analyses showed that

Table 2

Linear mixed-effect models with random intercept for participants and trials testing the effect of Campbellian environmental attitude on relative dwell time on carbon emissions controlling for attribute position (Model 1) and controlling for consistency, gender, age and perceived EU-ETS efficacy (Model 2). Coefficients represent unstandardized regression coefficients.

Predictors	Model 1			Model 2		
	<i>b</i>	95 % CI	<i>p</i>	<i>b</i>	95 % CI	<i>p</i>
(Intercept)	0.60	0.56–0.63	<.001	0.64	0.45–0.83	<.001
Campbellian attitude	0.03	0.00–0.05	.020	0.02	-0.00–0.05	.085
Attribute position	-0.16	-0.20–0.11	<.001	-0.15	-0.19–0.11	<.001
Decision consistency				0.03	-0.04–0.09	.394
Gender				0.05	0.01–0.10	.018
Age				-0.00	-0.01–0.00	.230
EU-ETS efficacy				0.00	-0.02–0.03	.718
Random Effects						
σ^2			0.03			0.03
τ_{00} part			0.01			0.01
τ_{00} trial			0.00			0.00
ICC			0.30			0.29
N_{part}			114			114
N_{trial}			36			36
Observations			3968			3968
Marginal/Conditional R^2			0.131/0.395			0.153/0.400
AIC/BIC/Log-likelihood			-2347/-2309/1180			-2350/-2287/1185
Likelihood Ratio Test						$\chi^2(4) = 10.77$, $p = .029$
Max VIF			1.01			1.12

Note. Attribute position was dummy-coded 1 for carbon emissions at the bottom, 0 for carbon emissions at the top. Gender was dummy-coded 1 for women, 0 for men.

women reported significantly stronger Campbellian attitudes score than men ($t(112) = 2.217, p = .029$), supporting this interpretation.

For models using SVS attitude as predictor, multicollinearity was similarly low (maximum VIF = 1.11). As with Campbellian attitude, model comparisons were inconclusive: difference in AIC values ($\Delta AIC = 4$) and LRT ($\chi^2(4) = 11.89, p = .018$) favored Model 2, while difference in BIC values ($\Delta BIC = 21$) continued to support the more parsimonious Model 1. Regardless of the model retained, SVS attitude was not a significant predictor of relative dwell time, thus failing to support H1 (Model 1: $b = 0.01, p = .255, 95\% \text{ CI } [-0.01, 0.02]$; Model 2: $b = 0.00, p = .735, 95\% \text{ CI } [-0.01, 0.02]$; see [Supplementary Table S2](#) and [Appendix 1](#)).

3.2.2. Environmental attitudes predicting first fixation on environmental attribute

To test the effect of environmental attitude on first fixations made towards the environmental attribute (H2), we fitted GLMMs including random intercepts for participants and trials, while controlling for attribute position in Model 3 and all covariates in Model 4. For models using Campbellian attitude as predictor, multicollinearity was low (maximum VIF of 1.12) and model comparison criteria consistently favored Model 3 ($\Delta AIC = 6$; $\Delta BIC = 30$; $\chi^2(4) = 2.92, p = .571$; see [Supplementary Table S3](#)). For models using SVS attitude as predictor, multicollinearity was also low (maximum VIF of 1.12) and model comparison criteria also consistently favored Model 3 ($\Delta AIC = 4$; $\Delta BIC = 30$; $\chi^2(4) = 3.41, p = .492$; see [Supplementary Table S3](#)). In these best-fitted models, H2 was not supported, as the position of the first fixation was not significantly predicted by the Campbellian attitude ($OR = 1.37, p = .114, 95\% \text{ CI } [0.93-2.02]$) or the SVS attitude scores ($OR = 1.09, p = .480, 95\% \text{ CI } [0.86-1.39]$).

We expected first fixations to reflect pre-existing attitudes only if participants were familiar with attribute locations. To facilitate this, attribute positions were kept constant across trials, enabling participants to anticipate where each attribute would appear. However, it is plausible that familiarity with attribute positions increased as trials progressed, reducing random scanning and potentially strengthening the relationship between attitudes and first fixation locations. To explore this, we conducted additional non-preregistered analyses by introducing the order of trials as a moderator in the best-fitted Models 3. For models using Campbellian attitude as predictor, they revealed a significant interaction between Campbellian attitude and order of trials ($OR = 1.18, p = .003, 95\% \text{ CI } [1.06-1.31]$), indicating that the effect of Campbellian attitude on first fixations indeed strengthened as participants completed more trials (see [Supplementary Table S5](#)). A simple slopes analysis revealed that the relationship between Campbellian attitude and first fixation on the environmental attribute was significant only in later stages of the task (e.g., at the 29th trial: $OR = 1.57, p = .010, 95\% \text{ CI } [1.09-2.23]$; see [Supplementary Fig. S3](#)). For models using SVS attitude as predictor, the interaction between SVS attitude and order of trials was not significant ($OR = 1.09, p = .118, 95\% \text{ CI } [0.98-1.21]$; see [Appendix 2](#) for more details).

3.2.3. Relative dwell time on environmental attribute predicting sustainable decision

To test the effect of relative dwell time on environmental attribute on pro-environmental decision (H3), we fitted GLMMs including random intercepts for participants and trials, while controlling for attribute position in Model 3 and all covariates in Model 4. Model comparison criteria consistently favored Model 4 ($\Delta AIC = 42, \Delta BIC = 18, \text{LRT: } \chi^2(4) = 50.63, p < .001$), and multicollinearity was not a concern (maximum VIF = 1.10). The hypothesis was not supported in the best-fitted model, as the relative dwell time on carbon emissions did not significantly predict decision ($OR = 1.12, p = .691, 95\% \text{ CI } [0.63-2.00]$; see [Table 3](#)). Notably, in contrast to our previous analyses predicting attention, we observed substantial random intercept variance for trials in the present models predicting decision-making. Model 4's high conditional R^2 value

indicated that 76.2 % of the variance in pro-environmental decision was explained by the model. However, the intraclass correlation coefficient ($ICC = 0.71$) suggested that most of this variability was attributable to variations in participants (variance = 3.88) and trials (variance = 4.16). The fixed effects alone only explained a small portion of the variance (marginal $R^2 = 0.181$).

To further investigate the role of trial-level variability on H3, we conducted non-preregistered analyses by fitting a GLMM including a random intercept for participants only, with relative dwell time and attribute position as fixed effects. In this simplified model, H3 was supported: higher relative dwell time on environmental attribute significantly predicted pro-environmental decisions ($OR = 1.74, p = .013, 95\% \text{ CI } [1.13-2.69]$; see [Table 3](#)). Since all trials followed the same design, differing only in the levels of monetary incentives and carbon emissions associated with each offer, these findings suggest that trial-specific factors (i.e., variations in bonus and carbon levels) contribute substantially to decision-making variability. Consistent with prior research ([Berger & Wyss, 2021](#); [Wyss et al., 2024](#)), this highlights the substantial role of contextual factors in shaping participants' decisions. While individual attention to carbon emissions may play a role, the specific monetary and environmental attributes of each offer appear to have a stronger influence on pro-environmental decision-making.

Building on these findings, we conducted further non-preregistered analyses to explore how trial-specific factors moderate the relationship between relative dwell time and pro-environmental decisions. To do so, we introduced bonus levels and carbon levels as moderators in the best-fitted Model 4. Both monetary incentives level ($OR = 0.88, p = .026, 95\% \text{ CI } [0.79-0.99]$) and carbon emissions level ($OR = 1.33, p < .001, 95\% \text{ CI } [1.19-1.50]$) significantly moderated this relationship (see [Supplementary Table S6](#)). A simple slopes analysis revealed that higher relative dwell time on carbon emissions predicted pro-environmental decision for offers with the lowest monetary incentives (5 CHF; $OR = 1.32, p = .020, 95\% \text{ CI } [1.05-1.66]$). However, the relationship was not significant at medium (14 CHF; $OR = 1.08, p = .220, 95\% \text{ CI } [0.95-1.23]$) or high monetary incentives (23 CHF; $OR = 0.88, p = .270, 95\% \text{ CI } [0.73-1.09]$). Similarly, the effect of relative dwell time on pro-environmental decisions was moderated by carbon emissions levels. At the highest level of carbon emissions (110 kg CO₂), relative dwell time on carbon emissions positively predicted pro-environmental decision ($OR = 1.73, p < .001, 95\% \text{ CI } [1.35-2.20]$). This relationship was not significant at the medium level of carbon emissions (65 kg CO₂, $OR = 1.08, p = .230, 95\% \text{ CI } [0.95-1.23]$) and became negative at the lowest level of carbon emissions (20 kg CO₂; $OR = 0.68, p < .001, 95\% \text{ CI } [0.55-0.84]$), indicating that higher relative fixation time on low carbon emissions was associated with a greater likelihood of accepting the related offers (see [Supplementary Fig. S4](#) and [Appendix 3](#) for more details).

3.2.4. Last fixation on environmental attribute predicting sustainable decision

To test the effect of last fixation on environmental attribute on pro-environmental decision (H4), we fitted GLMMs including random intercepts for participants and trials, while controlling for attribute position in Model 3 and all covariates in Model 4. Model comparison criteria consistently favored Model 4 ($\Delta AIC = 43, \Delta BIC = 18, \text{LRT: } \chi^2(4) = 50.73, p < .001$), and multicollinearity was not a concern (maximum VIF = 1.08). The hypothesis was not supported in the best-fitted model, as the last fixation on carbon emissions did not significantly predict decision ($OR = 0.98, p = .827, 95\% \text{ CI } [0.79-1.21]$; see [Table 4](#)). In this model, a high proportion of the total variance explained was attributable to random effects of participants and trials ($ICC = 0.71$).

As with H3, we conducted non-registered analyses to investigate the role of trial-level variability by using a GLMM including a random intercept for participants only, with last fixation and attribute position as fixed effects. In this simplified model, H4 was supported: a higher proportion of last fixations on carbon emissions significantly predicted

Table 3

Generalized linear mixed-effect models testing the effect of relative dwell time towards carbon emissions on pro-environmental decisions and controlling for attribute position, when accounting for variations between participants (simplified model), variations between participants and trials (Model 3) and when controlling for decision consistency, gender, age and perceived EU-ETS efficacy (Model 4). Coefficients represent odds ratio from logistic regressions.

Predictors	Simplified model			Model 3			Model 4		
	OR	95 % CI	p	OR	95 % CI	p	OR	95 % CI	p
(Intercept)	2.00	1.22–3.29	.006	4.41	1.61–12.12	.004	0.39	0.01–15.21	.618
Relative dwell time on carbon	1.74	1.13–2.69	.013	1.16	0.65–2.06	.624	1.12	0.63–2.00	.691
Attribute position	0.83	0.46–1.52	.546	0.69	0.27–1.81	.455	0.81	0.38–1.75	.599
Decision consistency							74.15	21.24–258.87	<.001
Gender							2.73	1.22–6.14	.015
Age							1.02	0.89–1.17	.792
EU-ETS efficacy							0.78	0.49–1.21	.267
Random Effects									
σ^2			3.29			3.29			3.29
τ_{00} part			2.39			6.33			3.88
τ_{00} trial						4.15			4.16
ICC			0.42			0.76			0.71
N_{part}			114			114			114
N_{trial}						36			36
Observations			3968			3968			3968
Marginal/Conditional R^2			0.005/0.424			0.003/0.762			0.181/0.762
AIC/BIC/Log-likelihood			4289/4314/-2140			2953/2985/-1472			2911/2967/-1446
Likelihood Ratio Test						$\chi^2(1) = 1337.12, p < .001$			$\chi^2(4) = 50.63, p < .001$
Max VIF			1.01			1.01			1.08

Note. Pro-environmental decision was dummy-coded 1 for declined offers, 0 for accepted offers. Attribute position was dummy-coded 1 for carbon emissions at the bottom, 0 for carbon emissions at the top. Gender was dummy-coded 1 for women, 0 for men.

Table 4

Generalized linear mixed-effect models testing the effect of last fixation towards carbon emissions on pro-environmental decisions and controlling for attribute position, when accounting for variations between participants (simplified model), variations between participants and trials (Model 3) and when controlling for decision consistency, gender, age and perceived EU-ETS efficacy (Model 4). Coefficients represent odds ratio from logistic regressions.

Predictors	Simplified model			Model 3			Model 4		
	OR	95 % CI	p	OR	95 % CI	p	OR	95 % CI	p
(Intercept)	2.49	1.62–3.85	<.001	4.88	1.87–12.69	.001	0.43	0.01–16.38	.651
Last fixation on carbon	1.24	1.05–1.46	.010	0.98	0.79–1.21	.863	0.98	0.79–1.21	.827
Attribute position	0.78	0.43–1.43	.428	0.68	0.26–1.76	.423	0.80	0.37–1.71	.559
Decision consistency							74.54	21.32–260.64	<.001
Gender							2.76	1.23–6.20	.014
Age							1.02	0.89–1.17	.800
EU-ETS efficacy							0.78	0.49–1.22	.268
Random Effects									
σ^2			3.29			3.29			3.29
τ_{00} part			2.43			6.36			3.89
τ_{00} trial						4.16			4.16
ICC			0.42			0.76			0.71
N_{part}			114			114			114
N_{trial}						36			36
Observations			3968			3968			3968
Marginal/Conditional R^2			0.005/0.427			0.003/0.762			0.181/0.763
AIC/BIC/Log-likelihood			4288/4313/-2140			2954/2985/-1472			2911/2967/-1446
Likelihood Ratio Test						$\chi^2(1) = 1336.48, p < .001$			$\chi^2(4) = 50.73, p < .001$
Max VIF			1.00			1.00			1.08

Note. Pro-environmental decision was dummy-coded 1 for declined offers, 0 for accepted offers. Last fixation was dummy-coded 1 on the environmental attribute, 0 if on the financial attribute. Attribute position was dummy-coded 1 for carbon emissions at the bottom, 0 for carbon emissions at the top. Gender was dummy-coded 1 for women, 0 for men.

pro-environmental decisions ($OR = 1.24, p = .010, 95\% \text{ CI } [1.05–1.46]$; see Table 4). These findings provide further evidence that trial-specific factors (i.e., variations in bonus and carbon levels) have a strong influence on pro-environmental decision-making variability.

Further exploratory, non-preregistered analyses were conducted to explore how trial-specific factors moderate the relationship between last fixation on carbon emissions and pro-environmental decisions. To do so, we introduced bonus levels and carbon levels as moderators in the best-fitted Model 4. The model revealed significant fixed effects of carbon emissions levels ($OR = 0.89, p = .038, 95\% \text{ CI } [0.80–0.99]$) and monetary incentives ($OR = 0.89, p = .038, 95\% \text{ CI } [0.80–0.99]$) on decision-making. However, there was no evidence of these levels moderating the relationship between last fixation location and decision (see

Supplementary Table S7 and Appendix 4 for more details).

3.3. Exploratory analyses on the moderating role of attentional processes on the relationship between environmental attitude and behavior

To test the moderating role of attentional processes on the attitude-behavior association, we performed mixed-effects models examining the interaction between Campbellian attitude and eye-tracking measures in predicting pro-environmental decisions. Campbellian attitude first emerged as a strong predictor of pro-environmental decisions across all models, with odds ratios ranging from 2.55 to 2.68 ($p < .001$). Regarding eye-tracking measures indexing the relative importance of the attributes, we found no moderating effect of first fixation ($OR = 0.92, p =$

Table 5

Generalized linear mixed-effect models testing the moderating role of number of gaze transitions between attributes on the relationship between Campbellian environmental attitude and pro-environmental decisions when controlling for attribute position (Model 3) and for decision time, decision consistency, gender, age and perceived EU-ETS efficacy (Model 4). Coefficients represent odds ratio from logistic regressions.

Predictors	Model 3			Model 4		
	OR	95 % CI	p	OR	95 % CI	p
(Intercept)	5.49	2.19–13.73	<.001	0.44	0.01–14.31	.642
Campbellian attitude	2.68	1.70–4.23	<.001	2.02	1.37–2.96	<.001
Number of transitions	0.86	0.77–0.96	.005	1.00	0.85–1.18	.982
Campbellian attitude * Number of transitions	0.85	0.76–0.94	.003	0.85	0.76–0.94	.002
Attribute position	0.57	0.23–1.40	.222	0.70	0.34–1.46	.340
Decision time				0.81	0.69–0.96	.013
Decision consistency				65.81	19.61–220.93	<.001
Gender				2.08	0.94–4.58	.069
Age				1.02	0.89–1.16	.796
EU-ETS efficacy				0.88	0.57–1.36	.571
Random Effects						
σ^2			3.29			3.29
τ_{00} part			5.52			3.52
τ_{00} trial			4.09			4.07
ICC			0.74			0.70
N part			114			114
N trial			36			36
Observations			3968			3968
Marginal/Conditional R^2			0.076/0.764			0.229/0.767
AIC/BIC/Log-likelihood			2926/2970/-1456			2883/2958/-1429
Likelihood Ratio Test						$\chi^2(5) = 53.25, p < .001$
Max VIF			1.02			2.45

Note. Pro-environmental decision was dummy-coded 1 for declined offers, 0 for accepted offers. Attribute position was dummy-coded 1 for carbon emissions at the bottom, 0 for carbon emissions at the top. Gender was dummy-coded 1 for women, 0 for men. Campbellian attitude, number of transitions and decision time were centered.

.363, 95 % CI [0.78–1.09]), relative dwell time ($OR = 0.99, p = .889$, 95 % CI [0.87–1.13]) or last fixation ($OR = 1.09, p = .121$, 95 % CI [0.98–1.21]) on the attitude-decision association. Similarly, among eye-tracking measures indexing the level of deliberation,³ mean fixation duration did not moderate the attitude-decision association ($OR = 1.02, p = .790$, 95 % CI [0.90–1.14]).

However, a significant interaction emerged between the number of gaze transitions and Campbellian attitude when predicting pro-environmental decisions ($OR = 0.85, p = .003$, 95 % CI [0.76–0.94], see Model 3 in Table 5). This indicates that the strength of the attitude-decision association varied depending on the level of deliberation, as reflected by the number of transitions between attributes. Post-hoc simple slopes analysis revealed that the attitude-decision association was stronger when participants made fewer gaze transitions between attributes. Specifically, the attitude-decision association was strongest at low levels of transitions (-1 SD: $OR = 2.39, p < .001$, 95 % CI [1.58–3.56]), followed by the mean level of transitions ($OR = 2.02, p < .001$, 95 % CI [1.37–2.96]), and weakest at high levels of transitions ($+1$ SD: $OR = 1.72, p = .010$, 95 % CI [1.16–2.53]; see Fig. 4A). These results remained consistent when controlling for all covariates, including decision time ($OR = 0.85, p = .002$, 95 % CI [0.76–0.94], see Model 4 in Table 5 and Appendix 5 for more details).

These findings suggest that a higher number of transitions, indicative of more deliberative processing, attenuates the relationship between attitude and decision. In other terms, the fewer transitions made by participants, the more likely they decided in alignment with their

attitudes. Fig. 4B further illustrates that the moderating effect of gaze transitions was particularly pronounced among participants with lower environmental attitudes. These participants acted less in accordance with their attitude - and thus more pro-environmentally - when they made more transitions between attributes.

4. Discussion

This study aimed to address the recent call from researchers for a deeper understanding of the cognitive processes involved in pro-environmental decision-making (Bamberg, 2013; Nielsen, 2017; Sawe & Chawla, 2021). Drawing on extensive literature on decision-making processes in social psychology, we examined how pre-existing environmental attitudes predict attention to relevant information for pro-environmental decision-making and how these attentional processes, in turn, are associated with decisions. We found some evidence consistent with these proposed relationships, though their expression appeared sensitive to the method of attitude measurement and features of the task environment (see a summary of findings in Supplementary Table S8).

Examining the link between environmental attitudes and attention, Model 1 showed that stronger environmental attitudes - when measured using the Campbell Paradigm - were associated with a greater focus on the environmental impact of choices, as reflected in increased relative dwell time on information about carbon emissions. In Model 2 including additional covariates, the effect was however attenuated and became marginal. Since the model comparison criteria did not provide consistent or conclusive evidence in favor of either model, we considered both models when interpreting our findings. The attenuated effect is potentially due to the confounding influence of gender, since women tend to report stronger environmental attitudes than men (as observed in this study and previous research; Mertens et al., 2021; Vicente-Molina et al., 2018; Xia & Li, 2023). Together, these findings suggest that personal ecological considerations may shape the weight given to environmental information during decision-making. This aligns with previous studies in social decision-making, which show that empathic concerns (Hu et al.,

³ To support the interpretation of our models, we ran preregistered control analyses testing the association between mean fixation duration (i.e., average of the duration of all fixations during the offer presentation in milliseconds) or number of gaze transitions between both attributes (i.e., moving one's gaze from one attribute to the other) and participants' retrospective self-reports of decision deliberation for each trial. As expected, Spearman's rank correlations showed positive correlations between number of transitions and decision deliberation ($p = 0.41, p < .001$) and between mean fixation duration and decision deliberation ($p = 0.20, p < .001$).

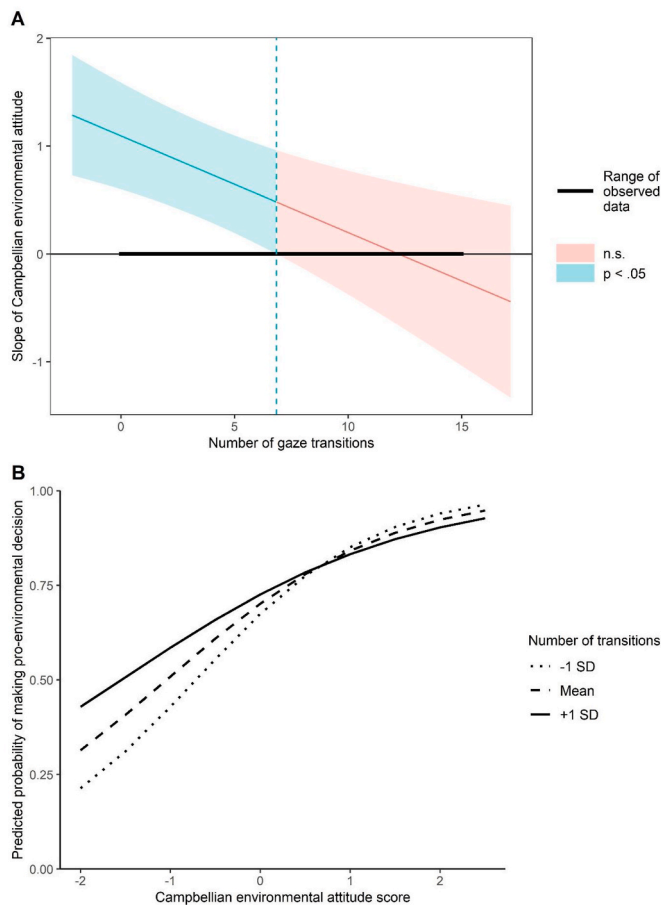


Fig. 4. (A) Johnson-Neyman plot illustrating the range of gaze transitions (0–15) where the effect of Campbellian environmental attitude on decision is statistically significant (blue shaded region). (B) Marginal effects estimates for the relationship between the Campbellian environmental attitude score (measured in logits) and the predicted probability of making a pro-environmental decision when participants made a low (–1 SD), intermediate (mean) or high (+1 SD) number of gaze transitions between attributes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2020) or social preferences (e.g., cooperative, competitive, or individualistic types; Fiedler et al., 2013; Jiang et al., 2016; Polonio et al., 2015) strongly affect the level of attention given to certain aspects of the decision-making context. Our findings also align with eye-tracking studies investigating attentional biases in the field of environmental psychology, showing that individuals with stronger environmental concerns allocate more visual attention to climate change stimuli (Carlson et al., 2019; Sollberger et al., 2017).

Our preregistered analyses indicated that environmental attitudes were overall not associated with initial fixations on carbon emissions. However, further exploration revealed that the relationship between the first fixation location and the Campbellian environmental attitudes progressively emerged as participants progress through the task and become familiar with the decision-making context. Our results are consistent with findings from Hu et al. (2020) who, using a similar counterbalancing procedure for attribute position in a decision task, observed that participants with stronger empathic concerns tended to direct their first fixations toward the victim's payoff information when the location of this information was kept constant within participants. Thus, as participants become familiar with the decision context, their environmental attitudes and pre-existing goals (e.g., reducing individual carbon emissions) may increasingly guide their initial focus of attention.

Overall, our data suggest that Campbellian environmental attitudes

are partly associated with attentional orientation, as evidenced by their link to increased proportion of fixation time on the environmental consequences of offers in Model 1 and a higher likelihood of directing the first fixation to this information only once participants became familiar with its location. However, when assessing environmental attitudes through biospheric values (i.e., SVS environmental attitude), no significant effects emerged. This distinction is particularly notable, as it highlights key differences in how environmental attitudes are conceptualized. While biospheric values capture broad environmental concern and moral commitment, the Campbell Paradigm offers a complementary perspective by incorporating personal cost considerations, which weaken the predictive power of conventional measures for trade-offs involving higher personal stakes. Indeed, Wyss et al. (2022) demonstrated a weaker association between environmental attitudes (assessed by the conventional New Ecological Paradigm) and pro-environmental behavior when personal costs are high, as seen in the present experiment involving substantial monetary stakes. In contrast, the Campbellian environmental attitude measure, which derives attitudes from the personal costs associated with a broad range of reported pro-environmental behaviors, may be better suited to our research question. This aligns with findings from Kaiser and Byrka (2014), who demonstrated that the Campbell measure, unlike conventional attitude measures, effectively distinguishes between individuals engaged in high-cost pro-environmental behaviors (such as a vegetarian diet) and those who are not. Additionally, its Rasch-based measurement approach provides a structured, unidimensional assessment, helping to reduce inconsistencies often observed in contemporary attitude research (Kaiser & Wilson, 2019). Although biospheric values remain valuable for understanding individuals' general environmental worldviews, they may be more influenced by social desirability or abstract ideals, which could explain differences in predictive power in high-cost decision contexts. Future research should explore how both approaches uniquely contribute to understanding real-world environmental decisions, where individuals must navigate personal benefits alongside environmental considerations.

Our study highlights the complex interplay between attention and pro-environmental decision-making. Although we observed that greater overall attention to environmental information (i.e., mean relative dwell time and mean proportion of last fixations on carbon emissions) predicted a higher proportion of pro-environmental decisions, this effect highly dissipated when accounting for trial-specific variables. This discrepancy may be explained by the fact that when exposed to the same set of decisions, participants who consistently allocate more attention to carbon emissions throughout the task tend to make more pro-environmental decisions overall. However, when focusing on individual trials, the specific monetary and carbon emission amounts emerge as the key predictors of decision outcomes. This indicates that while individual attention on decision outcomes is likely to play a role in shaping general decision patterns, the immediate context of each decision – particularly the personal costs and environmental stakes – exerts a more direct and substantial importance on each choice.

Further exploration of the interaction between those trial-specific factors and attention measures pointed to a moderating role of the amount of money in the relationship between relative dwell time and decision. Specifically, we observed that paying more attention to environmental consequences of an offer significantly predicts its rejection in trials with low financial incentives. Therefore, attention could be an important factor in low-cost decisions to protect the environment (i.e., offers with lower personal rewards) while more costly decisions could potentially rely on higher-level cognitive processes (e.g., working memory or self-control capacity; Baumgartner et al., 2019; Langenbach et al., 2020; Wyss et al., 2022). Additionally, we found that the proportion of fixation time towards the environmental consequences of an offer either predict its rejection or its acceptance depending on the amount of carbon emissions involved. When carbon emissions are relatively low (i.e., lower level than bonus amounts), longer attention to

this information is associated with a decreased likelihood of individuals rejecting the offer. We speculate that when participants pay more attention to low carbon levels, they may perceive the environmental consequences as minimal and thus feel justified to accept the monetary payment. Conversely, paying more attention to relatively high carbon emissions is related to an increased likelihood of individuals making pro-environmental decisions. Hence, a stronger attention to high carbon levels could motivate individuals to act responsibly by avoiding environmentally detrimental actions.

Finally, our exploratory analyses revealed that the number of gaze transitions significantly moderated the attitude-behavior relationship, even after accounting for decision time and other covariates. Specifically, the relationship between attitude and behavior was stronger when participants made fewer transitions between attributes (indicative of automatic and shallow information scanning) and weaker when participants frequently switched between the attributes (indicative of a more deliberative and deeper information search and evaluation). In other words, participants were more likely to act in accordance with their pre-existing environmental attitudes (e.g., accepting more environmentally harmful offers among individuals with lower environmental attitudes) when superficially scanning the attributes before deciding. Conversely, more thorough comparison and weighting of those attributes might have led participants to rather base their decisions on the specific characteristics of the offer. This effect appeared particularly pronounced for participants with low environmental attitudes, for whom greater deliberation on their decision (indexed by a higher number of transitions between attributes) is associated with more pro-environmental decisions. Thus, engaging in more conscious processing of offer attributes could prompt participants with lower environmental attitudes to act in a more environmentally friendly manner.

5. Conclusion

This study enhances our understanding of the complex interplay between environmental attitudes, attention, and pro-environmental decision-making. We found that attitudes derived from the Campbell paradigm are related to attentional processes, although this relationship may be partly attributable to gender-based differences in environmental attitudes and became more evident once participants had familiarized themselves with the spatial location of decision attributes. This relationship was not demonstrated when measuring environmental attitudes through biospheric values. The role of attention on decision-making is nuanced and context-dependent, with specific decision attributes and individual cognitive strategies being determinant. Additionally, exploring the role of attentional processes across a wider range of environmentally significant choices will be crucial for developing more effective strategies to promote sustainable behaviors. Altogether, this research contributes to the broader effort to promote individual actions against climate change by highlighting the role of environmental attitude, attentional processes, and decision-making context in shaping pro-environmental decisions.

CRedit authorship contribution statement

Zoé Bollen: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Annika Wyss:** Writing – original draft, Validation, Project administration, Conceptualization. **Emmanuel Guizar Rosales:** Writing – review & editing, Visualization, Validation, Formal analysis. **Zarah Le Houcq Corbi:** Conceptualization. **Daria Knoch:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Data Availability statement

The data and code are made publicly available on Open Science

Framework (<https://doi.org/10.17605/OSF.IO/WH7S6>) platform.

Declaration of interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvp.2025.102776>.

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